

NAME: _____

PERIOD: _____

Same Mean, Different Deadline Risk

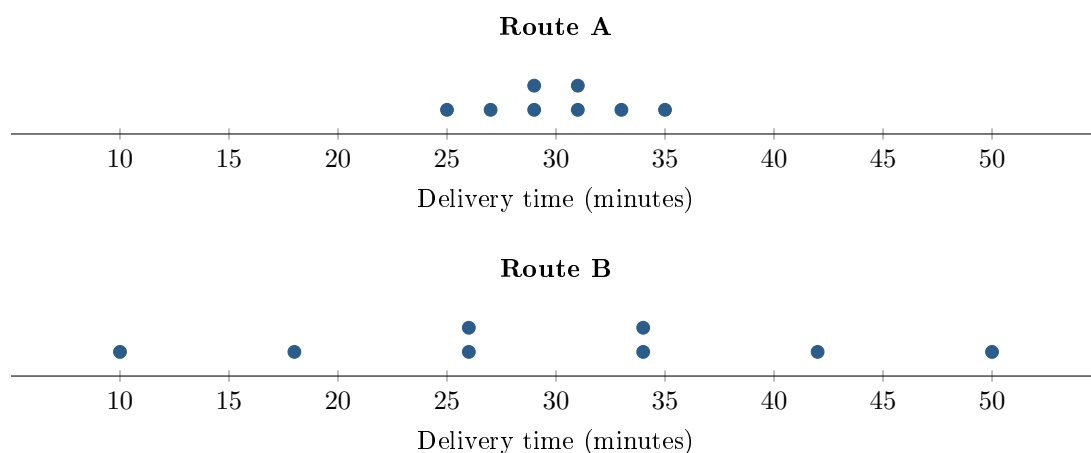
AP Statistics · Unit 2 · Topic 2.9

[STAR] THE PROBLEM

Suppose a dispatcher models delivery time using eight equally likely outcomes for each route, shown in the dotplots. An analyst reports that both routes have mean 30 minutes, with $\sigma_A = 3$ minutes and $\sigma_B = 12$ minutes.

The dispatcher says, *The routes have the same mean, so it does not matter which one a customer gets.* A customer has 40 minutes before an appointment and places a high cost on being late.

For the eight outcomes, the sums of squared distances from 30 are 72 for Route A and 1,152 for Route B.



FIRST TAKE — one sentence before you work the parts

Which route would you assign to this customer? Cite one number or dotplot feature.

[BOOK] MEAN AND STANDARD DEVIATION (keep on the page)

For a discrete random variable, $\mu_X = \sum xP(X = x)$ and $\sigma_X = \sqrt{\sum (x - \mu_X)^2 P(X = x)}$. For n equally likely outcomes, divide the sum of squared distances from the mean by n , then take the square root. The mean is a long-run average; the standard deviation describes a typical distance from it, in the variable's units. Add probabilities of outcomes that miss a deadline.

Work (a)–(c).

DOK 1 (a) Count the outcomes longer than 40 minutes for each route.

DOK 2 (b) Use the given sums of squared distances to verify both population standard deviations. Interpret what one of them means in minutes.

DOK 3 (c) Which route fits this customer's priority? Use both standard deviations and the chance of exceeding 40 minutes. Explain why the dispatcher's mean-only comparison can mislead this customer. Write 3–4 sentences.

Frame: Route _____ fits because its SD is _____ minutes and its chance of exceeding 40 minutes is _____.

Frame: The equal means describe _____, but they do not show _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.